

CO-3 CASE STUDY

STUDENT CASE STUDY EXECUTION REPORT

Department: FED

Subject: Data Mining

Academic Year: I-III Semester (2025-2026)

Faculty Name: Ms. G.Mamatha

Student Name: B. Devi Charitha

Roll Number: 2520030462

Branch: CSE

Course Outcome:

CO-3 PRISM Algorithm

Case Study Topic:

Student Performance Prediction using PRISM Algorithm

Work Done Summary:

In this case study, the PRISM Algorithm is used to analyze student academic data and predict performance levels. Student information such as attendance percentage, assignment marks, and internal examination scores are used to generate classification rules. These rules help classify students into performance categories such as Excellent, Good, Average, and Poor. The system assists faculty members in monitoring student progress and identifying students who require additional academic support.

Implementation Details:

1. Student academic data is collected from institutional records.
2. Attributes such as Attendance, Assignment Marks, and Internal Marks are selected for analysis.
3. Data preprocessing is performed to remove inconsistencies.
4. Performance categories are defined as Excellent, Good, Average, and Poor.
5. The PRISM Algorithm generates classification rules based on attribute values.
6. Rules are created to predict student performance accurately.
7. Students are classified according to generated rules.
8. Performance prediction results are analyzed and stored.
9. Faculty members can identify students requiring additional attention.
10. The system improves academic decision-making and performance monitoring.

Result : Screenshots / Output

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Student Performance Prediction using PRISM Algorithm

Enter Student Name: charitha
Enter Attendance Percentage: 95
Enter Internal Marks: 70

Generated Rule:
IF Attendance > 75 AND Internal Marks > 60
THEN Performance = Good

Prediction Result:
Student Name : charitha
Performance Category : Good

Classification Completed Successfully
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1. Student data was successfully classified using the PRISM Algorithm.
2. Classification rules were generated automatically.
3. Students were categorized as Excellent, Good, Average, or Poor.
4. Academic performance prediction was performed successfully.
5. The generated rules were easy to understand and interpret.
6. The system helped identify students needing academic support.
7. Prediction process improved educational decision-making.
8. Classification accuracy was achieved using rule-based prediction.

GitHub Link: <https://github.com/devicharithab-del/DSA-2-CO3>

Student Signature

Faculty Signature